TeachHub Lagos – Use Case Document

1. Introduction

This document describes the key interactions between users and the TeachHub Lagos platform. It identifies the main actors, their goals, and the steps involved in completing important platform activities.

The use cases are derived from the project's documented requirements and functionality. Detailed interaction steps and alternative scenarios are analytical elaborations for portfolio documentation.

2. System Overview

TeachHub Lagos is a centralised digital resource platform designed to help educators create, organise, manage, and share curriculum-aligned learning materials.

The platform provides different levels of access for teachers and administrators based on their roles and permissions.

3. Actors

Actor	Description
Teacher	Uses the platform to manage lessons and educational resources, create quizzes and flashcards, and control content visibility.
Administrator	Manages platform users, roles, permissions, and resources, and has access to platform-wide content.

4. Use Case Summary

ID	Use Case	Primary Actor
UC-01	Register an Account	Teacher
UC-02	Authenticate User	Teacher / Administrator
UC-03	Manage User Accounts	Administrator
UC-04	Manage Roles and Permissions	Administrator
UC-05	View Platform Content	Administrator
UC-06	Create and Manage Lessons	Teacher
UC-07	Attach Learning Materials	Teacher
UC-08	Create Quizzes	Teacher
UC-09	Create Flashcards	Teacher
UC-10	Preview Learning Content	Teacher
UC-11	Publish Learning Content	Teacher
UC-12	Manage Educational Resources	Teacher
UC-13	Delete Educational Resources	Administrator
UC-14	Set Content Visibility	Teacher

5. Detailed Use Cases

UC-01: Register an Account

Primary actor: Teacher

Goal: Create an account to access the platform.

Preconditions:

- The user can access the platform's registration interface.

Trigger: The user chooses to register.

Main flow:

1. The user opens the registration interface.
2. The user provides the required registration information.
3. The user submits the registration details.
4. The platform processes the registration request.
5. The platform creates the account if the information is accepted.

Alternative flow:

- If the submitted information is invalid or incomplete, the platform should request correction before registration can proceed.

Postcondition: A user account is created.

Note: The exact registration fields and account-verification process were not specified in the project documentation.

UC-02: Authenticate User

Primary actors: Teacher, Administrator

Goal: Access the platform using an authorised account.

Preconditions:

- The user has an account.

Trigger: The user attempts to sign in.

Main flow:

1. The user opens the login interface.
2. The user enters their credentials.
3. The platform validates the submitted credentials.
4. The platform grants access when authentication succeeds.
5. The platform presents the appropriate dashboard based on the user's role.

Alternative flow:

- If authentication fails, access is not granted.

Postcondition: The authenticated user gains access according to their assigned role.

UC-03: Manage User Accounts

Primary actor: Administrator

Goal: Manage accounts registered on the platform.

Preconditions:

- The administrator is authenticated.
- The administrator has the required user-management permissions.

Trigger: The administrator opens the user-management area.

Main flow:

1. The administrator accesses user management.
2. The platform displays available user-management information.
3. The administrator selects a user account.
4. The administrator performs an available management action, such as adding, modifying, or deleting an account.
5. The platform processes the action.

Alternative flow:

- If the selected account or requested action is invalid, the platform should not complete the operation.

Postcondition: The requested user-management action is processed.

UC-04: Manage Roles and Permissions

Primary actor: Administrator

Goal: Control user roles and access permissions.

Preconditions:

- The administrator is authenticated.
- The administrator has permission to manage roles.

Trigger: The administrator opens role or permission management.

Main flow:

1. The administrator accesses role-management functionality.
2. The administrator selects a user or role.
3. The administrator reviews the available role or permission settings.
4. The administrator makes the required change.
5. The platform processes the change.

Alternative flow:

- If the administrator does not have the required permission, the change should not be completed.

Postcondition: The relevant role or permission setting is updated.

UC-05: View Platform Content

Primary actor: Administrator

Goal: Access and review content available across the platform.

Preconditions:

- The administrator is authenticated.

Trigger: The administrator opens the platform content-management area.

Main flow:

1. The administrator accesses the content area.
2. The platform presents available platform content.
3. The administrator views the relevant resources.

Postcondition: The administrator can view platform content.

UC-06: Create and Manage Lessons

Primary actor: Teacher

Goal: Create and manage educational lessons.

Preconditions:

- The teacher is authenticated.
- The teacher has access to lesson-management functionality.

Trigger: The teacher chooses to create or manage a lesson.

Main flow:

1. The teacher opens the lesson-management area.
2. The teacher selects the option to create a lesson or manage an existing one.
3. The teacher enters or updates lesson information.
4. The teacher saves the changes.
5. The platform processes the lesson information.

Alternative flow:

- If required information is missing or invalid, the lesson may need to be corrected before the action can be completed.

Postcondition: The lesson is created or updated.

UC-07: Attach Learning Materials

Primary actor: Teacher

Goal: Add supporting educational materials to learning content.

Preconditions:

- The teacher is authenticated.
- The teacher is creating or managing relevant learning content.

Trigger: The teacher chooses to attach a learning resource.

Main flow:

1. The teacher opens the relevant lesson or content area.
2. The teacher selects the attachment option.
3. The teacher selects an available image, video, or worksheet.
4. The teacher submits the attachment.
5. The platform processes the resource.

Alternative flow:

- If the selected resource cannot be processed, the attachment may not be completed.

Postcondition: The learning content has the submitted attachment, provided the operation succeeds.

UC-08: Create Quizzes

Primary actor: Teacher

Goal: Create quiz content for educational use.

Preconditions:

- The teacher is authenticated.
- The teacher can access quiz-creation functionality.

Trigger: The teacher selects the quiz-creation option.

Main flow:

1. The teacher opens the quiz-creation interface.
2. The teacher enters the quiz information.
3. The teacher creates the quiz content.
4. The teacher saves the quiz.
5. The platform processes the quiz.

Postcondition: The quiz is created and available for subsequent content management.

Note: Specific question formats, scoring rules, and quiz-attempt workflows were not detailed in the available project requirements.

UC-09: Create Flashcards

Primary actor: Teacher

Goal: Create flashcards as learning resources.

Preconditions:

- The teacher is authenticated.
- The teacher can access flashcard-creation functionality.

Trigger: The teacher selects the flashcard-creation option.

Main flow:

1. The teacher opens the flashcard-creation interface.
2. The teacher enters the flashcard information.
3. The teacher creates the flashcard content.
4. The teacher saves the flashcards.
5. The platform processes the content.

Postcondition: The flashcards are created.

UC-10: Preview Learning Content

Primary actor: Teacher

Goal: Review learning content before publishing.

Preconditions:

- The teacher is authenticated.
- Learning content has been created or is available for review.

Trigger: The teacher selects the preview option.

Main flow:

1. The teacher opens the relevant content.
2. The teacher selects preview.
3. The platform displays the content for review.
4. The teacher checks the content before deciding whether to publish or make changes.

Postcondition: The teacher has reviewed the content.

UC-11: Publish Learning Content

Primary actor: Teacher

Goal: Make prepared learning content available through the platform.

Preconditions:

- The teacher is authenticated.
- The content is available for publishing.

Trigger: The teacher selects the publish option.

Main flow:

1. The teacher opens the content-management area.
2. The teacher selects the content to publish.
3. The teacher initiates the publishing action.
4. The platform processes the request.
5. The content is published if the operation succeeds.

Alternative flow:

- If the publishing operation fails, the content is not treated as successfully published.

Postcondition: The content is published.

UC-12: Manage Educational Resources

Primary actor: Teacher

Goal: Upload, edit, and view educational resources.

Preconditions:

- The teacher is authenticated.
- The teacher has access to the relevant resource-management functionality.

Trigger: The teacher opens the resource-management area.

Main flow:

1. The teacher accesses resource management.
2. The teacher selects an available resource-management action.
3. The teacher uploads a new resource, edits an existing resource, or views a resource.
4. The platform processes the requested action.

Alternative flow:

- If an upload or update cannot be completed, the platform should not treat the operation as successful.

Postcondition: The selected resource action is completed where permitted.

UC-13: Delete Educational Resources

Primary actor: Administrator

Goal: Remove educational resources from the platform.

Preconditions:

- The administrator is authenticated.
- The administrator has the required permissions.

Trigger: The administrator selects a resource for deletion.

Main flow:

1. The administrator accesses the relevant resource.
2. The administrator selects the delete action.
3. The platform processes the deletion request.
4. The resource is removed if the operation succeeds.

Alternative flow:

- Users without administrative permission cannot perform the resource-deletion action.

Postcondition: The selected resource is deleted.

UC-14: Set Content Visibility

Primary actor: Teacher

Goal: Control whether learning content is private or globally available.

Preconditions:

- The teacher is authenticated.
- The teacher has created or manages the relevant content.

Trigger: The teacher opens the content-visibility settings.

Main flow:

1. The teacher selects the relevant content.
2. The teacher opens the visibility settings.
3. The teacher chooses the available private or global visibility option.
4. The platform processes the setting.
5. The content is assigned the selected visibility where the operation succeeds.

Postcondition: The content has the selected visibility setting.

Note: The precise audience and access implications of the global setting should be confirmed against the implemented application.

6. Actor-to-Use-Case Mapping

Actor	Main responsibilities
Teacher	Register, authenticate, create and manage lessons, attach learning materials, create quizzes and flashcards, preview and publish content, manage resources, and set content visibility.
Administrator	Authenticate, manage user accounts, manage roles and permissions, view platform content, and delete educational resources.

7. UML Use-Case Diagram Guidance

A UML use-case diagram for TeachHub Lagos should show:

System boundary: TeachHub Lagos Platform.

Actors outside the boundary:

- Teacher
- Administrator

Teacher use cases inside the boundary:

- Register an Account
- Authenticate User
- Create and Manage Lessons
- Attach Learning Materials
- Create Quizzes
- Create Flashcards
- Preview Learning Content
- Publish Learning Content
- Manage Educational Resources
- Set Content Visibility

Administrator use cases inside the boundary:

- Authenticate User
- Manage User Accounts
- Manage Roles and Permissions
- View Platform Content
- Delete Educational Resources

Connect each actor to the use cases they perform. Authentication can be represented as a shared use case associated with both actors.

Avoid adding UML `include` or `extend` relationships unless the actual relationship between the use cases has been established. The diagram should communicate the documented system scope rather than introduce unsupported functionality.

8. Business Rules Reflected in the Use Cases

1. Users access platform functionality according to their assigned roles.
2. Teachers can create and manage lessons and educational resources.
3. Teachers can upload, edit, and view educational resources.
4. Administrators can manage user accounts, roles, and permissions.
5. Administrators can view platform content.
6. Resource deletion is restricted to administrators.
7. Learning content can be assigned private or global visibility.

9. Assumptions and Items Requiring Confirmation

The following details should be checked against the actual application before presenting these use cases as exact implementation behaviour:

- The complete registration fields and account-verification process.
- Whether users can reset forgotten passwords.
- The exact steps involved in editing and deleting user accounts.
- Whether lesson deletion is supported and which roles can perform it.
- The validation rules for uploaded files.
- The exact publishing workflow and whether publishing requires approval.
- The meaning and audience of global content visibility.
- Whether teachers can access resources created by other teachers.
- The exact quiz and flashcard creation fields and behaviour.

These items are intentionally not presented as confirmed features.

10. Business Analyst Contribution

The use-case documentation demonstrates how the Business Analyst translated platform requirements into actor-centred system interactions.

This involved:

- Identifying the primary actors and their responsibilities.
- Mapping functional requirements to user goals.
- Defining preconditions, triggers, workflows, and postconditions.
- Identifying alternative and exception scenarios.
- Highlighting permission boundaries between teachers and administrators.
- Documenting assumptions and questions requiring stakeholder confirmation.
- Providing a foundation for UML use-case modelling and functional testing.

11. Conclusion

The use cases provide a structured view of how teachers and administrators interact with the TeachHub Lagos platform.

They complement the business requirements, functional requirements, and user stories by describing the expected interactions associated with the platform's core capabilities.

Together, these documents demonstrate the Business Analyst's role in translating an identified educational problem into documented, testable system requirements.

Portfolio documentation note: This document is a reconstructed, sanitised representation of the project requirements for professional demonstration. The detailed flows are not a claim that every step or exception was implemented exactly as written.